

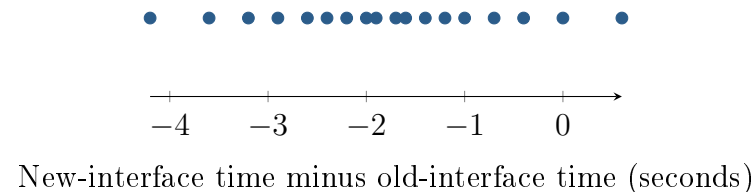
# Two Measurements per Technician

Unit 4 · Topic 4.4 · THE PROBLEM

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows  $d = \text{new time} - \text{old time}$ . Researchers ask whether the new interface reduces the population mean time.

**Nia:** “The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail.”

**Colin:** “There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative.”



## FIRST TAKE — one sentence before you work the parts

Would you compare each technician’s two times or treat them as separate people? Explain in one sentence.

**DOK 1** (a) Define  $\mu_d$  in context.

**DOK 2** (b) Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

**DOK 3** (c) In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

**Self-paced bonus problem.** Work (a)–(c) from this sheet; turn it in whenever you finish.